Section 2.2 Areas & area ratios

面积与面积比

[AIME-2002-II-Q04] | Part A (Q1-Q5)

In a regular hexagonal cluster of side-1 regular hexagon cells, there are n cells along each of the six outer sides. Only its outermost cell ring is paved; the enclosed interior is a garden. At $n = 202$, write garden area as $m\sqrt{3}/2$. Find $m \bmod 1000$.

用边长一的正六边形单元形成规则六边形簇，每条外侧边上有 n 个单元。仅最外一圈铺成小路，内部未铺部分为花园。取 $n = 202$，若花园面积为 $m\sqrt{3}/2$，求 m 除以 1000 的余数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2003-I-Q02] | Part A (Q1-Q5)

Concentric circles have radii $1, 2, \ldots, 100$. Starting with a red center disk, color successive regions alternately red and green. Write the green area divided by the radius-100 disk area as m/n reduced. Find $m+n$.

同心圆半径依次为 1 至 100。中心圆盘染红，向外各环红绿交替。绿色总面积占最大圆盘面积的比约分为 m/n，求 $m+n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2004-I-Q04] | Part A (Q1-Q5)

In a side-2 square, consider length-2 segments with endpoints on adjacent sides. Their midpoint arcs enclose a central region. If its area rounded to the nearest hundredth is k, find $100k$.

边长为二的正方形中，取端点位于相邻两边且长为二的线段。它们的中点弧围出中央区域，面积四舍五入到百分位为 k，求 $100k$。

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Working / 解题过程

Section 2.2 Areas & area ratios

面积与面积比

[AIME-2004-II-Q01] | Part A (Q1-Q5)

A chord is perpendicular to a radius at its midpoint. The larger-to-smaller disk-region ratio is $(a\pi + b\sqrt{c})/(d\pi - e\sqrt{f})$, with positive integers, $\gcd(a, e) = 1$, and squarefree c, f. Find $abcdef \bmod 1000$.

一条弦在半径中点处与半径垂直。它分圆盘所得大、小区域面积比写为 $(a\pi + b\sqrt{c})/(d\pi - e\sqrt{f})$，参数为正整数，$a, e$ 互质，c, f 无平方因子。求乘积 $abcdef$ 模 1000。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-II-Q01] | Part A (Q1-Q5)

A convex equilateral hexagon $ABCDEF$ has right angles at A, D and four other equal angles. Its area is $2116(1+\sqrt{2})$. Find its side length.

凸等边六边形 $ABCDEF$ 在 A, D 为直角，其余四角相等，面积为 $2116(1+\sqrt{2})$。求边长。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2008-I-Q02] | Part A (Q1-Q5)

An isosceles triangle has as its base one side of a side-ten square, with apex on the same side as the square interior. Its overlap with the square has area eighty. Find its altitude to that base.

等腰三角形以边长十的正方形一边为底，顶点在正方形内部所在一侧，两图重叠面积八十。求三角形底边上的高。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2014-II-Q03] | Part A (Q1-Q5)

A hinged rectangle has side lengths $a, 36$, with hinges at its corners and at the midpoints of its length-36 sides. Press the a-sides symmetrically inward, keeping them parallel and their midpoints perpendicularly aligned, to form a convex hexagon. When the separation is 24, its area equals the rectangle area. Find a^2.

边长为 $a, 36$ 的矩形在各顶点及三十六长边中点设铰链。将两条 a 边对称向内压，保持平行且两中点连线垂直于它们，形成凸六边形。两边相距二十四时，面积与原矩形相同。求 a^2。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2019-I-Q03] | Part A (Q1-Q5)

Triangle PQR has $PR = 15, QR = 20, PQ = 25$. On PQ mark A, B, on QR mark C, D, and on PR mark E, F, with $PA = QB = QC = RD = RE = PF = 5$. Find $[ABCDEF]$.

三角形 PQR 边长为十五、二十、二十五。A, B 在 PQ，C, D 在 QR，E, F 在 PR，满足 $PA = QB = QC = RD = RE = PF = 5$。求六边形面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2021-II-Q02] | Part A (Q1-Q5)

An equilateral triangle ABC has side 840. On the perpendicular ray from B toward the side containing A, choose D. The line through D parallel to BC meets AB, AC at E, F. Choose G on the same line beyond F from E, so AFG is isosceles and $[AFG] : [BED] = 8 : 9$. Find AF.

等边三角形 ABC 边长为 840。由 B 向含 A 一侧作 BC 的垂线并取点 D。过 D 作 BC 平行线交 AB, AC 于 E, F。在该线上取 G，使 F 位于 E, G 之间，AFG 为等腰三角形，且 $[AFG] : [BED] = 8 : 9$。求 AF。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2025-I-Q02] | Part A (Q1-Q5)

On triangle sides, A, D, E, B occur in order with successive lengths $4, 16, 8$, and A, F, G, C occur in order with lengths $13, 52, 26$. Reflect D across point F to obtain M, and G across point E to obtain N. If $[DEGF] = 288$, find the area of polygon $AFNBCEM$.

三角形两边上，A, D, E, B 依次分段为 $4, 16, 8$，A, F, G, C 依次分段为 $13, 52, 26$。将 D 关于点 F 对称得 M，将 G 关于点 E 对称得 N。若 $DEGF$ 面积为 288，求多边形 $AFNBCEM$ 的面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2000-II-Q06] | Part B (Q6-Q10)

A trapezoid has bases differing by 100. Its leg-midpoint segment divides the area in ratio $2:3$. A parallel section of length x divides the area equally. Find $\lfloor x^2/100 \rfloor$.

梯形两底差为 100。两腰中点连线将面积分成 $2:3$。另有一条平行两底、长为 x 的截线将面积平分，求 $\lfloor x^2/100 \rfloor$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2001-I-Q09] | Part B (Q6-Q10)

In a triangle with $AB = 13, BC = 15, CA = 17$, take $D \in AB, E \in BC, F \in CA$ with $AD/AB = p, BE/BC = q, CF/CA = r > 0$. Given $p + q + r = 2/3$ and $p^2 + q^2 + r^2 = 2/5$, write $[DEF]/[ABC] = m/n$ reduced and find $m + n$.

三角形边长 $AB = 13, BC = 15, CA = 17$，边上点满足 $AD/AB = p, BE/BC = q, CF/CA = r > 0$。已知 $p + q + r = 2/3, p^2 + q^2 + r^2 = 2/5$。若面积比 $[DEF]/[ABC] = m/n$ 已约分，求 $m + n$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2003-II-Q06] | Part B (Q6-Q10)

Triangle ABC has sides $13, 14, 15$. Rotate it 180° about its centroid to obtain a second triangle. Find the area of the union of the two filled triangles.

边长为 $13, 14, 15$ 的三角形绕重心旋转 180° 得第二个三角形，求两个实心三角形并集面积。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-I-Q07] | Part B (Q6-Q10)

An angle is cut by seven equally spaced parallel cross-lines numbered outward from its vertex. Region A is the initial triangle; B, C, D are strips between lines 2–3, 4–5, and 6–7. Given $[C]/[B] = 11/5$, find $[D]/[A]$.

一个角被七条等距平行截线依次截开。A 为顶点到第一线的三角形；B, C, D 分别为第二至三、第四至五、第六至七线间的带状区。若面积比 $[C]/[B] = 11/5$，求 $[D]/[A]$。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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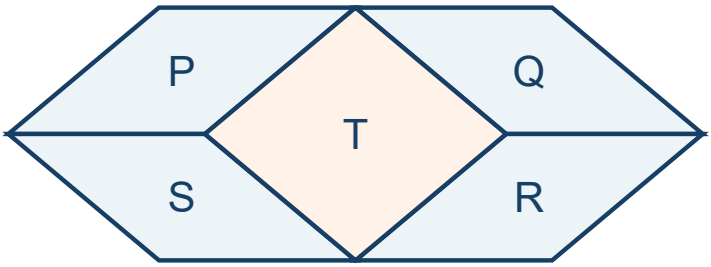
Section 2.2 Areas & area ratios

面积与面积比

[AIME-2006-I-Q08] | Part B (Q6-Q10)

A hexagon is tiled by five equal-side rhombi as in the companion diagram: a central rhombus T and four congruent outer rhombi P, Q, R, S. At the top vertex of T, the incident angles of P, T, Q form a straight angle; likewise at the bottom. Each outer rhombus has area $\sqrt{2006}$. How many positive integer areas can T have?

如配图，六边形被五个等边长菱形铺满：中央为 T，四周 P, Q, R, S 全等。T 顶点处 P, T, Q 的角拼成平角，底点处同理。每个外围菱形面积为 $\sqrt{2006}$。中央面积有多少个正整数可能值?



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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2011-I-Q08] | Part B (Q6-Q10)

A triangular sheet has sides $23, 27, 30$. At each vertex cut off a corner by a fold line parallel to the opposite side, with the three corner flaps nonoverlapping along the original edges. Fold all three corners down by 90° to form a table with horizontal top and all three vertices on the floor. Its greatest height is $k\sqrt{m}/n$, with coprime k, n and squarefree m. Find $k + m + n$.

一张三角形纸的边长为 $23, 27, 30$。在三顶点各作平行于对边的折线，三个角片在原各边上不重叠。将三角片向下折九十度，使桌面水平、三个原顶点落地。最大高度为 $k\sqrt{m}/n$，k, n 互质、m 无平方因子，求 $k + m + n$。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2013-II-Q10] | Part B (Q6-Q10)

For a circle of radius $\sqrt{13}$, an exterior point A is $4+\sqrt{13}$ from its center. Let B be its nearest circle point. A secant through A meets the circle at K, L. The greatest area of BKL is $(a-b\sqrt{c})/d$, with coprime a, d and squarefree c. Find $a+b+c+d$.

半径 $\sqrt{13}$ 的圆外有点 A，距圆心 $4+\sqrt{13}$。B 为圆上离 A 最近点。过 A 的割线交圆于 K, L。若 BKL 最大面积为 $(a-b\sqrt{c})/d$，a, d 互质、c 无平方因子，求四者和。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2007-I-Q12] | Part C (Q11-Q15)

Isosceles triangle ABC has $A=(0,0), B=(20,0)$, $AC=BC$, C in the first quadrant and $\angle BAC=75^\circ$. Rotate about A until C reaches the positive y axis. The overlap area is $p\sqrt{2}+q\sqrt{3}+r\sqrt{6}+s$ with integer coefficients. Find $(p-q+r-s)/2$.

等腰三角形 ABC 中 $A=(0,0), B=(20,0), AC=BC$，C 在第一象限，$\angle BAC=75^\circ$。绕 A 逆时针旋转至 C 落在正 y 轴，重叠面积为 $p\sqrt{2}+q\sqrt{3}+r\sqrt{6}+s$，求 $(p-q+r-s)/2$。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2009-I-Q11] | Part C (Q11-Q15)

Choose two distinct nonnegative lattice points P, Q on $41x + y = 2009$ and form triangle OPQ with the origin. Count distinct triangles of positive integer area.

从直线 $41x + y = 2009$ 上非负整数格点中选不同两点 P, Q，与原点组成三角形，求正整数面积的不同三角形数。

Mathematical statement reworded for this bilingual edition. 原数学条件的双语重述。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2010-I-Q13] | Part C (Q11-Q15)

A rectangle $ABCD$ and a semicircle on diameter AB have disjoint interiors. A line meets the semicircle at N, AB at U, and CD at T, all distinct, dividing their combined region in area ratio $1:2$. Given $AU=84, UB=168, AN=126$, write $DA=m\sqrt{n}$ with squarefree n and find $m+n$.

矩形 $ABCD$ 与以 AB 为直径的半圆内部不重叠。一条直线依次交半圆弧、AB, CD 于不同点 N, U, T，将合并区域面积分成 $1:2$。已知 $AU=84, UB=168, AN=126$，若 $DA=m\sqrt{n}$ 且 n 无平方因子，求 $m+n$。

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Section 2.2 Areas & area ratios

面积与面积比

[AIME-2025-II-Q14] | Part C (Q11-Q15)

Right triangle ABC has right angle A and hypotenuse $BC = 38$. Interior points K, L satisfy $AK = AL = BK = CL = KL = 14$. If quadrilateral $BKLC$ has area $n\sqrt{3}$, find n.

直角三角形 ABC 在 A 处为直角，斜边 $BC = 38$。内部点 K, L 满足 $AK = AL = BK = CL = KL = 14$。若四边形 $BKLC$ 面积为 $n\sqrt{3}$，求 n。

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